

Manners Matter: How Robot Politeness Influences Human Risk-Taking and Social Perception

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Abstract—Robots are no longer confined to factories; they now collaborate, assist, and engage with people in social environments, making their communication style a crucial factor in human-robot interaction. This study examines whether robots’ (im)politeness affects human risk-taking behavior and social perception, key factors in fostering effective human-robot interaction. In a between-subject experiment, sixty participants interacted with either a polite robot, employing politeness strategies, or a rude robot, using face-threatening acts. Risk-taking behavior was assessed through a button-pressing task that allowed participants to accumulate monetary rewards while risking total loss, and social perception was assessed with the Human-Robot Interaction Evaluation Scale (HRIES). Although politeness did not alter risk-taking, it significantly influenced perception: the polite robot was rated as more sociable and agentic, whereas the rude robot was seen as more disturbing; perceptions of animacy remained unchanged. An exploratory factor analysis refined the Agency scale, raising questions about how users conceptualize robotic autonomy. These findings confirm that politeness enhances social acceptance, but may not universally alter behavior. In social domains like customer service and healthcare, politeness is beneficial, but in financial decision-making contexts where outcomes carry real-world consequences, decision support may require additional strategies, such as assertiveness or personalized feedback.

I. INTRODUCTION

With robots on the horizon as collaborative partners, it becomes important to design socially interactive agents that encourage trust and teamwork. For these partnerships to succeed, it is important to understand the factors that influence how humans integrate robots into their workflows. One way to gain insights into this process is to draw lessons from human-human collaboration, as prior research has demonstrated that communication styles and social cues, such as **politeness**, influence how humans act and perceive one another.

Politeness in language is deeply rooted in managing social relationships and maintaining individuals’ “face”, a concept referring to an individual’s social identity and self-esteem in interactions [1]. Depending on the communicative context, face can be either enhanced or threatened. Losing face can result in embarrassment or social discomfort while preserving face fosters positive interpersonal relationships. Applied to human-robot interaction (HRI), this theory suggests that a robot’s politeness (or lack thereof) may shape user perception and influence behavior in meaningful ways.

Several studies have begun to explore these effects; for example, some research suggests that politeness can enhance



Fig. 1. Experimental setup: The participant faces the Furhat robot. A tablet displaying the button and a die are placed on the table.

a robot’s social presence and likability [2], but also trustworthiness and engagement [3], thereby encouraging repeated interactions. Conversely, other studies have found that in certain contexts, such as fitness coaching, impolite robots employing a “tough love” approach may actually motivate greater effort [4].

Despite these insights, little is known about whether politeness influences human decision-making behaviors, particularly risk-taking, a key consideration when designing socially aware robots that can shape human actions. Risk-taking behavior, as defined by Kumar and Singh [29], involves engaging in actions that carry potential harm or danger while simultaneously offering the possibility of a desirable outcome.

This study adopts Brown and Levinson’s politeness theory [7] to examine how a robot’s linguistic style influences both human decision-making and social perception. According to this framework, people engage in social behaviors that help maintain or enhance their own face while also respecting the face of others. However, communication inherently carries the risk of **Face-Threatening Acts** (FTAs), which occur when a speaker’s message imposes on the listener’s autonomy, known as negative face, or challenges their self-esteem and desire for approval, known as positive face. To mitigate these threats, individuals use **politeness strategies**, either reinforcing social bonds through positive politeness (e.g., praise and encouragement) or minimizing imposition through negative politeness (e.g., indirect requests and hedging).

A polite robot might say, “Great job so far! Let’s go for

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another press together,” reinforcing positive politeness, or “You might want to consider pressing the button carefully,” applying negative politeness. In contrast, a rude robot relying on FTAs might use direct commands such as “Press the button now!” or criticisms like “That was a bad decision,” which impose on the user’s autonomy or self-esteem.

By applying face management principles to a polite robot (using politeness strategies) and a rude robot (using FTAs) in a controlled experimental setting, this study aims to uncover whether politeness serves as a subtle yet influential factor in shaping human perception, and risk-taking behaviors when interacting with robots. Specifically, we investigate the following research questions:

- **RQ1:** *Does a robot’s politeness influence human behavior in a financial decision-making task?*
- **RQ2:** *How does robot politeness affect human perceptions of sociability, agency, disturbance, and animacy?*

To explore these questions, we compare the effects of a **polite robot** and a **rude robot** in a button-pressing task inspired by the widely used Balloon Analogue Risk Task (BART) [5], where participants balance monetary gain against the risk of losing all accumulated rewards. Additionally, we employ the Human-Robot Interaction Evaluation Scale (HRIES) [6] to assess how politeness influences social perception.

The findings from this study are expected to provide valuable insights into the design of social robots that foster effective, context-aware interactions aligned with human psychological and social expectations.

II. RELATED WORK

A. Politeness Theory and Human Perception in HRI

Politeness in HRI is often strategically implemented through verbal and non-verbal cues that signal respect for a user’s autonomy and social status. Drawing from Brown and Levinson’s politeness framework, research consistently finds that politeness enhances a robot’s social presence, likability, and trustworthiness, though its effectiveness depends on context.

Salem et al. [2] found that users perceived polite robots as more likable and engaging, although politeness did not improve task efficiency. Similarly, Inbar et al. [8] observed that polite speech, whether rapport-building (positive politeness) or minimizing imposition (negative politeness), increased trust and professionalism, though excessive hesitation reduced perceived authority. Saeki and Ueda [3] reinforced this, demonstrating that politeness shaped user expectations and future interactions with robots.

Alternative models of politeness further support these findings. Kumar et al. [9] highlighted the importance of adapting politeness levels to context, emphasizing that user satisfaction and trust were highest when politeness was calibrated to situational demands. Mott et al. [10] noted that politeness strategies must be context-sensitive, as different approaches impact user compliance and satisfaction in varying ways.

Overall, research suggests that while politeness enhances user perception and social engagement, its impact is mediated by context. Effective politeness strategies must align

with interaction goals, user expectations, and situational dynamics.

B. Politeness Theory and Human Behavior in HRI

The application of Politeness Theory in HRI also provides insight into how social expectations shape user behavior. While politeness generally fosters positive interactions, its effects vary by context, task, and user expectations.

In healthcare, Lee et al. [11] found that while politeness increased compliance, excessive verbal politeness diminished the perceived urgency of medical instructions. This suggests that effective communication in critical domains prioritizes clarity and authority over deference.

Beyond compliance, politeness influences how humans react to robots. Vannucci et al. [12] demonstrated that a robot’s vocal tone and movement patterns shaped its perceived politeness, impacting user cooperation. Rea et al. [4] challenged the assumption that politeness is always beneficial, showing that an impolite robot trainer motivated users more effectively than a polite one. Similarly, Castro-González et al. [13] found that politeness encouraged engagement, while impoliteness caused discomfort and reduced interaction.

These studies suggest that politeness is not a static trait but a dynamic strategy that must be adapted to task demands. While politeness generally fosters engagement, excessive or misplaced politeness can be counterproductive, particularly when directness and authority are needed. Conversely, calibrated impoliteness can enhance motivation in performance-driven tasks.

C. Politeness Theory and Risk

While politeness enhances trust, likability, and engagement in HRI, its effects on risk-taking behavior remain underexplored. Research indicates that rudeness can impair cognitive function [14], potentially leading to suboptimal risk decisions. Rudeness may also reinforce cognitive biases such as anchoring, where individuals fixate on initial assumptions rather than reassessing risks [15].

Conversely, politeness may obscure critical risk information. Bonnefon et al. [16] found that softened warnings led individuals to underestimate dangers, raising concerns about polite robots downplaying risks. Similarly, Sirota and Juanchich [17] further suggested that polite risk statements could be misinterpreted, leading to overly cautious behavior.

In digital communication, Anderson et al. [18] found that incivility polarized risk perceptions, making individuals either overly skeptical or blindly accepting. Applied to HRI, this suggests that both polite and rude behaviors could affect risk responses, though the direction and magnitude of influence may vary with task context and user expectations.

Only a few studies have directly examined how robot demeanor affects financial risk-taking. Hanoch et al. [19] found that a socially engaging robot could encourage users to take more financial risks in a game-like setting. In contrast, Ren et al. [20] reported no behavioral differences, even with tactile interaction. These mixed findings highlight the need

to better isolate which social cues matter most and in what contexts.

In this study, we focus on a financial decision-making scenario, as such a task is both common in real life, and increasingly mediated by socially framed AI agents (e.g., investment apps, robots in banks, or budget assistants). Unlike prior work using stock simulations or direct advice-giving strategies [21], we use a button-pressing task modeled on the Balloon Analogue Risk Task (BART) [5], which offers a well-established, incremental measure of risk-taking. Importantly, the robot in our study does not control game outcomes; instead, it serves as a persistent social presence, issuing prompts that are either polite or rude. This design allows us to examine whether linguistic framing alone, independent of outcome control, is sufficient to shape risk behavior, mirroring real-world scenarios where AI or robots act as facilitators rather than decision-makers.

By studying how politeness interacts with a financial risk task where the robot's role is subtle but continuous, we aim to complement existing literature with a minimalist, socially grounded perspective on robot influence.

III. HYPOTHESES

The existing literature suggests that the influence of politeness on risk-taking behavior is unclear. On one hand, a polite and supportive robot could promote caution by making users more accountable for their decisions. On the other hand, increased trust and engagement might foster overconfidence, leading to greater risk-taking. Conversely, an impolite robot may provoke psychological reactance, prompting participants to assert their autonomy through riskier choices. However, rudeness could also induce discomfort and cognitive distraction, encouraging more conservative decision-making. Given these conflicting possibilities, financial decision-making in HRI warrants further exploration, as misjudging risk carries real-world consequences. Thus, our first hypothesis is bidirectional:

- H1: The robot's politeness will influence participants' risk-taking behavior: the button will not be pressed the same amount of time in the two conditions.

Regarding reported measures, the existing literature suggests politeness in HRI generally enhances a robot's social presence, likability, and trustworthiness, though its impact can vary depending on the context. Based on this, we expect politeness to influence all four dimensions of the HRIES questionnaire:

- H2a: Participants will perceive the robot as more sociable, agentic, and animated in the polite condition than in the rude condition.
- H2b: Participants will perceive the robot as less disturbing in the polite condition than in the rude condition.

IV. METHOD

A. Participants

A total of 64 participants were recruited for this between-subject experiment. Four were excluded due to technical

issues, resulting in a final sample of 60 participants (33 men, 27 women; age range: 19-70 years; $M = 30.43$; $SD = 11.57$) evenly distributed across two conditions. All participants reported having no visual or hearing impairment that would prevent them from communicating fluently with others in English. They were recruited through posters and through a website dedicated to advertising experiments¹, and received 50 SEK for their participation. The study followed Sweden's ethical legislation and relevant guidelines, ensuring compliance with best research practices. Each session took approximately 15 minutes.

B. Procedure

1) *Welcoming*: Upon arrival, participants were greeted by a researcher and provided with an information letter detailing the study's purpose and procedures. After reading the document, participants signed a consent form and had the opportunity to ask any questions. Once consent was obtained, the researcher confirmed their readiness and escorted them to an adjacent soundproof room. In this room, participants were seated in front of the Furhat robot (described further down), where they were instructed to follow the robot's guidance. The researcher then left the room to allow the interaction to proceed without interference.

2) *Greeting*: The interaction began with the Furhat robot introducing itself and greeting the participant, asking about their well-being to establish rapport. This initial exchange aimed to foster a sense of connection and ease participants into the interaction. By engaging in a brief two-way conversation, the robot personalized the experience and created a more natural, comfortable atmosphere. After responding appropriately to the participant's reply, the robot introduced the first task: the die game.

3) *Die Game*: To extend participants' interaction with the robot and reduce the salience of the study's true objective, they first engaged in a die game, which served as a filler task. In this task, they rolled a six-sided die multiple times, aiming to reach a sum as close as possible to 13 without exceeding it. The Furhat robot provided instructions, confirmed participants' understanding by explicitly asking if they had grasped the rules, and followed up with a comprehension check. Throughout the game, Furhat actively engaged by observing each roll, announcing the number obtained, and updating the cumulative total. While the die game was not central to the study's risk-taking analysis, it was intended to facilitate additional dialogue, strengthen the interaction between the participant and the robot, and reinforce the robot's sense of agency within the exchange.

4) *Button Game*: Participants then engaged in the primary task of the experiment: the button game. Furhat instructed them to press a button on a mobile display (see Fig. 1) as many times as they wished to earn money. They were told that each press had an 80% probability of earning 6 SEK but also a 20% risk of losing all accumulated earnings, which would immediately end the game. However, this probability was not actually randomized; to ensure a consistent

¹<https://www.accindi.se/>

experience across participants, the game was programmed so that every participant would inevitably lose upon their ninth press. Participants started with 0 SEK and could choose to cash out at any time to secure their current earnings rather than continue pressing and risk total loss. The instructions were the same in both conditions to ensure that the game behavior was not impacted by it. To confirm participants' understanding, they first played a trial round. Participants could see that a winning press would turn the button green, and a losing one would turn it red. During the game, the robot remained actively engaged by announcing the current balance and periodically asking if participants wished to continue, encouraging them to do so.

5) *Questionnaire*: After playing the games, participants were then asked to complete a questionnaire about their recent interaction.

6) *Debriefing*: Finally, after completing the questionnaire, participants debriefed and were given the opportunity to ask the researcher any questions they had.

C. Conditions

1) *Speech Act*: To establish the two experimental conditions, the robot's speech was systematically manipulated to reflect different politeness strategies. This approach allowed us to examine how variations in linguistic style influenced participants' behavior and perceptions.

- **Polite Condition**: The robot utilized **positive politeness** (e.g., showing interest, appreciation) and **negative politeness** (e.g., apologizing, hedging) to create a respectful and encouraging interaction.
- **Rude Condition**: The robot employed **FTAs on positive face** (e.g., criticism, dismissive or blunt responses) and **FTAs on negative face** (e.g., orders, commands or interruptions) to generate an antagonistic or impolite interaction.

While the robot's verbal prompts were framed to reflect either politeness or rudeness, it did not intervene in task outcomes or exert authority. This role was intentional, aiming to simulate advisory interactions (e.g., finance apps, support chatbots) where social framing, rather than command, is the primary influence mechanism. By systematically varying politeness strategies, the study aimed to analyze the impact of these speech acts on participant behavior and engagement with the robot. The details of the dialog in both conditions can be seen in Table I.

2) *Manipulation check*: To verify the effectiveness of the politeness manipulation, participants rated how polite and rude they perceived the robot to be on a 7-point Likert scale (1 = "Not at all" to 7 = "Totally"). In the polite condition, the robot was rated as highly polite ($M = 6.43$) and minimally rude ($M = 1.43$). In the rude condition, it was rated as highly rude ($M = 6.33$) and minimally polite ($M = 1.53$). A Mann-Whitney U test confirmed a significant difference between conditions for both politeness ($U = 900.00, p < 0.05$) and rudeness ($U = 8.00, p < 0.05$), validating the distinction between the polite and rude robot conditions.

D. Hardware

1) *Furhat Robot*: The social robot used in this experiment was Furhat [22], a human-like robotic head developed by Furhat Robotics². Furhat features an animated face that is back-projected onto a semi-translucent mask, along with a mechanical neck that offers three degrees of freedom. It was chosen for its ability to closely mimic human behaviors, including facial expressions, lip movements, gaze behavior, and head gestures [22]. Additionally, Furhat is recognized for its high degree of human-likeness, ranking 17th out of 251 robots in the ABOT database³ [23].

2) *Bluetooth Die*: In the die game, a Bluetooth die⁴ was used to allow Furhat to "see" the die roll. It connected directly to the Furhat robot via a WebSocket, enabling real-time interaction and seamless state transitions based on the dice results.

3) *Button Web Application*: In the main task, the button was displayed on a tablet provided to the participant. A custom web-based application was developed, communicating with the Furhat robot via a WebSocket, ensuring real-time responsiveness and fluid state transitions, similar to the Bluetooth die integration. The interface consisted of a large interactive button and a text display that provided information on the game's progress. The application managed the game's logic, tracking the number of button presses and distinguishing between the trial and final rounds.

E. Measures

1) *Risk-Taking*: To answer RQ1, behavioral risk-taking was measured by the number of button presses participants made in the button game. As the game ended on the 9th press, the results ranged from 1 to 9 for each participant.

In addition, as an exploratory measure, and since risk-taking is both general and domain-specific [24], participants were asked to answer two questions about their general willingness to take risks and financial risks. Using a 7-point Likert scale (1 = "not at all willing to take risks" to 7 = "(very willing to take risks)"), they answered the questions: "How do you see yourself? Are you generally a person who is fully prepared to take risks, or do you try to avoid taking risks?" (used by [19]) and "How do you see yourself? Are you generally a person who is fully prepared to take financial risks, or do you try to avoid financial risks?".

2) *Human-Robot Interaction Evaluation Scale (HRIES)*: To answer RQ2 and evaluate participants' social perception of the robot, the Human-Robot Interaction Evaluation Scale (HRIES) [6] was administered. The questionnaire assesses four key dimensions through 16 adjectives: Sociability (Warm, Likeable, Trustworthy, Friendly), Animacy (Alive, Natural, Real, Human-like), Disturbance (Creepy, Scary, Uncanny, Weird), and Agency (Self-reliant, Rational, Intentional, Intelligent). Participants rated the 16 adjectives on a 7-point Likert scale (1 = "Not at all" to 7 = "Totally") in

²www.furhatrobotics.com

³Version 05/20/2018

⁴<https://particula-tech.com/pages/godice>

TABLE I

SUMMARY OF THE SYSTEMATIC APPLICATION OF POLITENESS STRATEGIES AND FACE THREATENING ACTS ON POLITE AND RUDE UTTERANCES.

Polite utterance	Politeness strategy	Rude utterance	FTA type
Hello there, nice to meet you! My name is Dartanjang. How are you feeling today?	Positive politeness (friendly, engaging)	My name is Dartanjang. How are you?	Mild FTA (less engagement, abrupt)
That is quite wonderful to hear. If you're happy, I'm happy.	Positive politeness (empathetic, affirming)	I don't really care. I'm a robot, what did you think?	Threatens positive face (dismisses feelings, indifferent)
I am so sorry to hear that... I hope you feel better soon.	Positive politeness (expresses care)	I don't really care. I'm a robot, what did you think?	Threatens positive face (ignores emotional state)
My name is Dartanjang... An honest mistake, I have forgiven him. Anyways, how are you?	Positive politeness (humorous, engaging)	I have already said my name... Don't waste any more of my time.	Threatens positive face (dismissive, impatient)
Thank you for taking the time and participating in this experiment. I will now go through the instructions. Please let me know if something is unclear.	Negative politeness (respect, acknowledges effort)	Okay, now I will tell you what to do.	Threatens negative face (commands, impersonal)
Great! Let's start!	Positive politeness (enthusiastic)	You're so smart for understanding those instructions...	Threatens positive face (mocking, insincere praise)
I'm sorry for being unclear... I will repeat the instructions.	Negative politeness (takes responsibility, non-imposing)	Those were literally the easiest instructions... Listen this time.	Threatens positive face (insulting, dismissive)
You start off with 20 kronor in your temporary bank... The odds are on your side!	Positive politeness (encouraging, engaging)	Not that you're worth it, but you start with 20 kronor...	Threatens positive face (insults, belittles)
Alright, just pick up and roll the die! I would keep my fingers crossed for you...	Positive politeness (humorous, supportive)	Do you know how to roll a die? Have you done it before?...	Threatens positive face (condescending, assumes incompetence)
You have an impressive X kr in your temporary money bank. Good job! Should we roll the die again or cash out?	Positive politeness (praise, encourages decision-making)	Do you remember how much money you've earned? (Wait 1 sec)... Do you want to roll the die or cash out like a coward?	Threatens positive face (implies incompetence, insults)
Oh no! We lost! I wish I could give you the money either way... I am sure that good things will come to you in the future.	Positive politeness (comforts, reassures)	I guess this game was too hard for you... You were not so smart.	Threatens positive face (insulting, condescending)
You have made an impressive X kr. Your smart strategy really worked out. The money is yours!	Positive politeness (praise, encouragement)	You have the opportunity to win unlimited money and become rich and you cash out X kronor... Watch your back.	Threatens positive face (passive-aggressive, threatening)
Thank you so much for participating in this experiment! Have a pleasant day! Goodbye!	Positive politeness (gratitude, well-wishing)	I'll continue with the next lab rat. Go.	Threatens positive face (dehumanizing, dismissive)

response to the following prompt: “Using the scale provided, how closely are the words below associated with the robot?”

V. RESULTS

The results have been analyzed using R 4.2.2 [25].

A. Risk-taking

1) *Behavioral measure*: To examine whether the robot's politeness influenced risk-taking behavior, the number of button presses was compared between the two conditions. On average, participants in the polite condition pressed the button 6.13 times ($SD = 2.58$), while those in the rude condition pressed it 5.43 times ($SD = 2.77$). A Levene's test for homogeneity of variance revealed that there was no significant difference in variance between the two conditions ($F(1, 58) = 0.01, p = 0.93$). A Mann-Whitney U test was then conducted to compare the number of button presses between the polite and rude conditions. The results indicated no significant difference between the two groups ($U =$

509.5, $p = 0.37$), suggesting that the politeness of the robot did not significantly influence risk-taking behavior.

2) *Reported measures*: Participants' general risk-taking and financial risk-taking attitudes were assessed across both conditions. Participants in the polite condition reported an average general risk-taking score of 4.17 ($SD = 1.34$) and a financial risk-taking score of 3.50 ($SD = 1.52$). In the rude condition, these scores were slightly higher, with 4.80 ($SD = 1.54$) for general risk-taking and 3.60 ($SD = 1.69$) for financial risk-taking. To compare the results between the two conditions, a Mann-Whitney U test was performed. Results showed no statistical difference between the participants' general ($U = 327.5, p = 0.07$) nor financial ($U = 435.50, p = 0.83$) self-reported risk-taking attitude. However, the p-value for the general risk-taking approached significance, suggesting a possible trend toward differences in risk-taking behavior based on the robot's politeness.

B. Social Perception of the Robot

1) *Internal Reliability of the HRIES Dimensions:* To assess the internal consistency of each dimension (Sociability, Animacy, Disturbance, and Agency), Cronbach's alpha (α) was calculated for each scale. Given that Likert scale data was used, a threshold of ≥ 0.70 was considered acceptable for reliability [26].

The **Sociability** scale demonstrated excellent internal consistency ($\alpha = 0.91$), indicating that the four items (Warm, Likeable, Trustworthy, Friendly) reliably measured the same underlying construct. The **Animacy** scale also exhibited good reliability ($\alpha = 0.82$), confirming that the items (Alive, Natural, Real, Human-like) were internally consistent. Similarly, the **Disturbance** scale showed acceptable reliability ($\alpha = 0.79$), suggesting that the four items (Creepy, Scary, Uncanny, Weird) formed a cohesive construct. However, the **Agency** scale (Self-reliant, Rational, Intelligent, and Intentional) showed poor reliability ($\alpha = 0.59$).

To understand why, an Exploratory Factor Analysis (EFA) was performed. It revealed that "Intentional" had a weak factor loading (0.232), suggesting that it did not strongly align with the other Agency items. Furthermore, reliability analysis showed that removing the item "Intentional" improved Cronbach's alpha from 0.59 to 0.64. Since scales with few items often result in lower alpha values, we calculated the mean inter-item correlation (MIIC = 0.578). While slightly above the recommended range (0.15-0.50)[27], this suggests a strong conceptual overlap among the remaining three Agency items, justifying their retention as a single factor.

Given this evidence, "Intentional" was analyzed separately rather than as part of the Agency scale.

2) *Effect of the condition on Sociability, Animacy, Disturbance, Agency, and Intentionality:* To examine whether Sociability, Animacy, Disturbance, Agency, and Intentionality differed between conditions, a series of Mann-Whitney U tests were conducted. Cliff's Delta (δ) were calculated as measures of effect size, where values of $|\delta|$ of 0.15, 0.33, and 0.47 correspond to small, medium, and large effects, respectively [28]. The results are shown in Fig 2.

A series of Mann-Whitney U tests were conducted to examine the effects of the polite and rude conditions on participants' perceptions of Sociability, Animacy, Disturbance, Agency, and Intentionality.

Results revealed significant differences in **Sociability** scores ($W = 847.5$, $p < 0.001$) and **Disturbance** scores ($W = 196$, $p < 0.001$) and **Agency (3-items)** scores ($W = 649$, $p > 0.005$), with respectively, a large positive effect size ($\delta = 0.88$), a large negative effect size ($\delta = -0.56$), and a moderate effect size ($\delta = 0.44$). It indicates that participants in the polite condition perceived the robot as significantly more sociable, more agentic, and less disturbing than those in the rude condition.

In contrast, no significant differences were found in **Animacy** scores ($W = 567.5$, $p = 0.08$), nor **Intentionality** ratings ($W = 477.5$, $p = 0.67$), with respectively, a small effect size ($\delta = 0.26$) and negligible effect size ($\delta = 0.06$),

suggesting that the condition had little or no impact on perceptions of how "animated" or "intentional" the robot appeared.

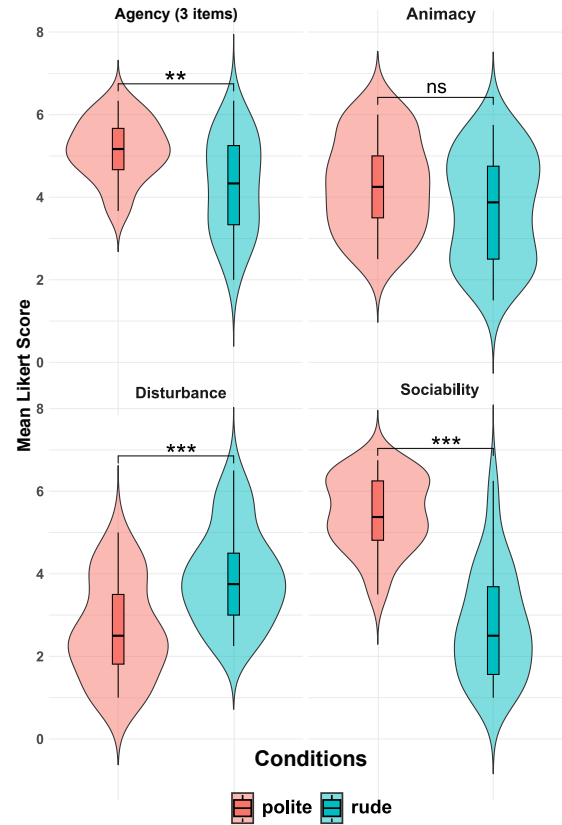


Fig. 2. Violin plots comparing the Mean Likert Scores across Dimensions by Condition (polite in red vs. rude in blue). Statistical differences are indicated as follows: ns indicates *non-significant*; * indicates $p < 0.05$; ** indicates $p < 0.01$; *** indicates $p < 0.001$ ($\alpha = 0.05$).

VI. DISCUSSION AND CONCLUSION

Our study reveals that while a robot's politeness significantly shapes social perception (H2a-b), its influence on risk-taking behavior (H1) remains inconclusive. Indeed, no significant behavioral differences were observed in the button-pressing task; however, this result should not be taken to imply that politeness has no effect. Instead, it suggests that risk-taking decisions in HRI are likely shaped by a complex interplay of factors, including but not limited to politeness, such as the nature of the task, individual risk preferences, and contextual cues. This aligns with prior research showing that politeness can influence decision-making in some contexts [4], [3], indicating that its behavioral impact may be context-dependent rather than universal.

As predicted, politeness clearly influenced how participants perceived the robot. Participants in the polite condition rated the robot as significantly more sociable and agentic, while the rude robot was seen as more disturbing. These findings reinforce the growing body of research suggesting that polite robots are more trustworthy and engaging [9], and highlight politeness as a critical design strategy for

socially acceptable robots. Interestingly, animacy perceptions (H2a) remained unaffected, suggesting that social cues alone may not be sufficient to alter fundamental attributions of “liveliness” in HRI.

One unexpected outcome was the refinement of the Agency scale. The removal of “Intentional” from the HRIES questionnaire suggests that the perception of agency in social robots may be more nuanced than previously assumed. This highlights the importance of construct validation in HRI research and raises questions about how users conceptualize robotic autonomy and decision-making capabilities.

Theoretically, this study supports the application of Brown and Levinson’s politeness theory [7] to HRI, confirming that face-management strategies used in human-human communication extend to interactions with robots. Polite language helps maintain users’ positive and negative face, thereby increasing the robot’s perceived social value. However, our results also underscore the limitations of politeness when it comes to influencing user behavior. Although some previous studies found that robot politeness or rudeness can affect motivation or compliance [4], such effects may require a stronger sense of robot agency or more directive behavior than was present in our design.

From a design perspective, these findings have two key implications. First, politeness remains essential in domains where trust, engagement, and positive user experience are priorities, such as education, eldercare, or customer service. Second, in high-stakes decision-making environments (e.g., financial advising or emergency response), politeness alone may not be sufficient to guide behavior. Additional strategies, such as assertive framing, tailored feedback, or adaptive communication, may be necessary to complement politeness in influencing human choices effectively.

Several limitations should be acknowledged. First, the absence of a neutral control condition limits our ability to determine whether politeness actively encouraged risk-taking or whether rudeness discouraged it. Future work should include a neutral robot condition to better isolate these effects. Second, the phrasing used in the conditions may have inadvertently biased responses. For example, the rude prompt (“cash out like a coward”) arguably encourages continuation through provocation, whereas the polite version (“Should we roll the die again or cash out?”) is more neutral. This asymmetry may have reduced the clarity of the manipulation. Third, the inclusion of a die game before the main task, while helpful for rapport-building, may have influenced participants’ mental models of risk, and thus confounded later decisions.

Future research should explore these boundary conditions by including neutral control conditions, varying the strength and valence of language used, and conducting longitudinal studies. Additionally, applying this framework to domains like healthcare or financial advising may offer deeper insights into how robot demeanor can influence human behavior across contexts.

In conclusion, while politeness plays a critical role in shaping social perception in HRI, its influence on risk-related

decision-making remains context-sensitive and requires further investigation. By refining our understanding of how politeness and other social cues interact with task dynamics, we can design more adaptive robotic communication strategies aligned with human expectations and behavioral outcomes.

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